

$$x_i(a_i, a_2, a_3, t)$$

5. Sea el movimiento $x_i = \left(1 + \frac{t}{k}\right) a_i$, donde k es una constante, t es el tiempo y a_i es la componente i -ésima de la posición inicial.

a. Determine la ecuación diferencial para la variación de densidad en el tiempo, a partir de la ecuación de conservación de masa.

b. Si la densidad en $t=0$ es ρ_0 , calcular la variación de la densidad con el tiempo.

Ayuda: Usar coordenadas espaciales para calcular $\text{div } v$ en la primera parte. En la segunda parte, usar separación de variables para resolver la ecuación diferencial.

$$x_i = \left(1 + \frac{t}{k}\right) a_i \Rightarrow$$

$$a_i = \frac{x_i}{1 + \frac{t}{k}} = \frac{k x_i}{k + t}$$

$$x_i = \left(1 + \frac{t}{k}\right) a_i \rightarrow \text{label}$$

$$v_i = \frac{dx_i}{dt} = \frac{1}{k} a_i$$

$$\Rightarrow \underline{v} \cdot \underline{v} = \frac{\partial v_i(a(x), t)}{\partial x_i} = \frac{\partial v_i}{\partial a_j} \cdot \frac{\partial a_j}{\partial x_i} = \frac{\delta_{ij}}{k} \cdot \frac{k}{k+t} \cdot \frac{a_j}{k+t}$$

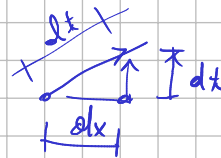
$$= \frac{3}{k+t}$$

pág 217 del libro:

▲ (10.5-4)

$$\frac{D\rho}{Dt} + \rho \frac{\partial v_j}{\partial x_j} \Big|_{t=\text{cte}} = 0$$

$\underline{v} \cdot \underline{v} \rightarrow \text{div}(\underline{v})$



$$\frac{D\rho}{Dt} + \rho \frac{\partial v_j}{\partial x_j} = 0.$$

Ec. Dif. de conservación de masa

$$\frac{D}{Dt}(x_i(t), t) = \dot{v}_i = \frac{\partial v_i(x(t), t)}{\partial t} \Big|_{x=\text{fijo}} \cdot \frac{\partial t}{\partial t} + \frac{\partial v_i}{\partial x_j} \Big|_{t=\text{fijo}} \cdot \frac{\partial x_j}{\partial t}$$

velocidad v_j

$$v_i(a_i, t) \wedge a_i = \frac{k x_i}{k+t} \Rightarrow v_i(x_i, t) = \frac{1}{k} \frac{k x_i}{k+t} = \frac{x_i}{k+t}$$

$$\underline{v} \cdot \underline{v} = \frac{\partial v_i(x, t)}{\partial x_i} \Big|_{t=\text{fijo}} = \frac{\partial}{\partial x_i} \left(\frac{x_i}{k+t} \right) = \frac{\delta_{ii}}{k+t} = \frac{3}{k+t}$$

$$\frac{D\rho}{Dt} + \rho \cdot \frac{3}{k+t} = 0$$

$$\frac{d\rho}{dt} + \rho \cdot \frac{3}{k+t} = 0$$

$\rho(t)$

$$\frac{d\rho}{dt} = -\rho \cdot \frac{3}{k+t}$$

$$\frac{1}{\rho} d\rho = -\frac{3}{k+t} dt$$

$$\int \frac{1}{\rho} d\rho = \int \frac{-3}{k+t} dt$$

$$u = k+t \Rightarrow du = dt$$

$$\ln \rho = -3 \int \frac{1}{u} du = -3 \ln u + C = -3 \cdot \ln(k+t) + C$$

1^{ra} versión

$$p = e^{-3 \ln(k+t) + C}$$

$$p(t=0) = e^{-3 \cdot \ln(k+0)} + C = p_0 \Rightarrow C = p_0 - e^{-3 \ln(k)}$$

$$p = e^{-3 \ln(k+t)} + p_0 - e^{-3 \ln(k)}$$

2^{da} versión

$$p = e^{-3 \ln(k+t) + C} = e^{-3 \ln(k+t)} \cdot e^C$$

$$p = C^* \cdot e^{-3 \ln(k+t)}$$

$$p(t=0) = C^* \cdot e^{-3 \cdot \ln(k+0)} = p_0$$

$$C^* = p_0 / e^{-3 \cdot \ln(k)} = p_0 \cdot e^{3 \ln(k)}$$

$$p = p_0 \cdot e^{3 \ln k} \cdot e^{-3 \ln(k+t)} = p_0 \cdot \frac{e^{3 \ln k}}{e^{3 \ln(k+t)}} = p_0 \cdot \frac{e^{\ln k^3}}{e^{\ln(k+t)^3}} = p_0 \cdot \frac{k^3}{(k+t)^3}$$

3^{ra} versión

$$\ln p = -3 \cdot \ln(k+t) + C$$

$$\ln p + \ln(k+t)^3 = C$$

$$\ln(p \cdot (k+t)^3) = C \Rightarrow p \cdot (k+t)^3 = e^C \Rightarrow p = \frac{C^*}{(k+t)^3}$$

$$p(t=0) = \frac{C^*}{(k+0)^3} = p_0 \Rightarrow C^* = p_0 k^3$$

$$p(a, t) = \frac{p_0 k^3}{(k+t)^3} = \frac{p_0}{\left(1 + \frac{t}{k}\right)^3}$$

Verificación:

$$\frac{dp}{dt} + p \cdot \frac{3}{k+t} = 3 \frac{p_0}{\left(1 + \frac{t}{k}\right)^4} \cdot \frac{1}{k} + \frac{p_0}{\left(1 + \frac{t}{k}\right)^3} \cdot \frac{3}{k + \frac{t}{k}} = 0$$

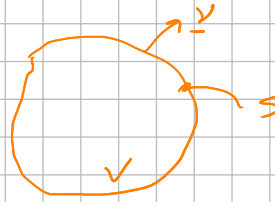
3. Sea V el volumen encerrado por una superficie S de normal saliente unitaria $\mathbf{n}$, y sean $\phi = \phi(x_1, x_2, x_3)$ y $\psi = \psi(x_1, x_2, x_3)$ funciones escalares de las coordenadas x_i . Usando el teorema de Green, demostrar:

$$\int_S (\phi \nabla \psi - \psi \nabla \phi) \cdot \mathbf{n} \, dS = \int_V (\phi \Delta \psi - \psi \Delta \phi) \, dV$$

donde $\Delta \psi = \psi_{,ii}$ es el laplaciano de la función ψ .

→ No se puede aplicar directamente el teorema de Gauss

$$\int_V \frac{\partial}{\partial x_i} (A_{jkl} \dots) \, dV = \int_S \nu_i A_{jkl} \dots \, dS,$$



$$C_j = \int_V \frac{\partial \pi_{ij}}{\partial x_i} \, dV = \int_S \pi_{ij} \nu_i \, dS$$

$$B_{ji} = \int_V \frac{\partial u_{ij}}{\partial x_i} \, dV = \int_S u_{ij} \nu_i \, dS$$

$$\int_S \phi \nabla \psi \cdot \mathbf{n} = \int_S \phi \cdot \frac{\partial \psi}{\partial x_i} \cdot \nu_i$$

$$\begin{aligned} \int_V \frac{\partial}{\partial x_i} \left(\phi \frac{\partial \psi}{\partial x_i} \right) &= \int_V \frac{\partial \phi}{\partial x_i} \frac{\partial \psi}{\partial x_i} + \int_V \phi \frac{\partial^2 \psi}{\partial x_i \partial x_i} \\ - \int_V \frac{\partial}{\partial x_i} \left(\psi \frac{\partial \phi}{\partial x_i} \right) &= - \int_V \frac{\partial \psi}{\partial x_i} \frac{\partial \phi}{\partial x_i} - \int_V \psi \frac{\partial^2 \phi}{\partial x_i \partial x_i} \end{aligned}$$

$$\int_V \frac{\partial}{\partial x_i} \left(\phi \frac{\partial \psi}{\partial x_i} \right) - \int_V \frac{\partial}{\partial x_i} \left(\psi \frac{\partial \phi}{\partial x_i} \right) = \int_V (\phi \Delta \psi - \psi \Delta \phi)$$

$$\int_S \phi \frac{\partial \psi}{\partial x_i} \nu_i - \int_S \psi \frac{\partial \phi}{\partial x_i} \nu_i = \int_V (\phi \Delta \psi - \psi \Delta \phi)$$

$$\int_S \phi \nabla \psi \cdot \mathbf{n} - \int_S \psi \nabla \phi \cdot \mathbf{n} = \int_V (\phi \Delta \psi - \psi \Delta \phi)$$

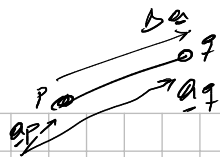
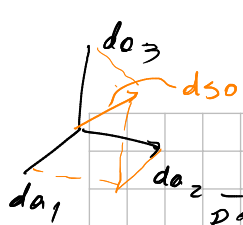
$$\int_S \phi \nabla \psi \cdot \mathbf{n} - \psi \nabla \phi \cdot \mathbf{n} = \dots$$

Variación 2

$$\int_S (\phi \nabla \psi) \cdot \mathbf{n} = \int_V \frac{\partial}{\partial x_i} \left(\phi \frac{\partial \psi}{\partial x_i} \right) = \int_V \frac{\partial \phi}{\partial x_i} \frac{\partial \psi}{\partial x_i} + \int_V \phi \frac{\partial^2 \psi}{\partial x_i \partial x_i}$$

$$- \int_S \psi \nabla \phi \cdot \mathbf{n} = - \int_V \frac{\partial}{\partial x_i} \left(\psi \frac{\partial \phi}{\partial x_i} \right) = - \int_V \frac{\partial \psi}{\partial x_i} \frac{\partial \phi}{\partial x_i} - \int_V \psi \frac{\partial^2 \phi}{\partial x_i \partial x_i}$$

$$\int_S \phi \nabla \psi \cdot \mathbf{n} - \int_S \psi \nabla \phi \cdot \mathbf{n} = \int_V \phi \frac{\partial^2 \psi}{\partial x_i \partial x_i} - \int_V \psi \frac{\partial^2 \phi}{\partial x_i \partial x_i}$$

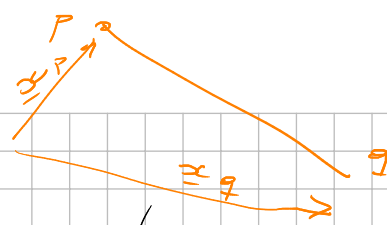


$$\vec{pq} = l = \|\Delta \underline{a}\| = \|\underline{da}\|$$

$$ds_0 = \|\underline{da}\| = \sqrt{da_i \cdot da_i}$$

$$ds^2 - ds_0^2 = da_i dx_i - da_i da_i$$

=



$$ds = \|\underline{dx}\| = \sqrt{dx_i \cdot dx_i}$$

$$x(a_1, a_2, a_3)$$

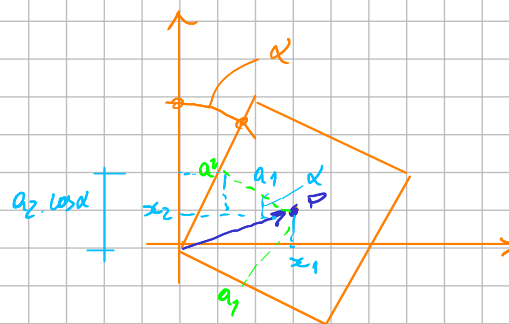
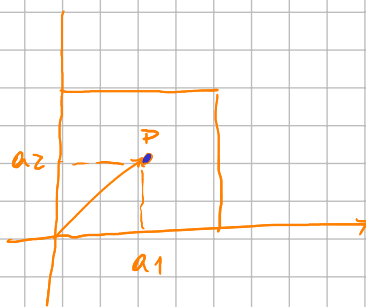
$$a_i(x_1, x_2, x_3)$$





$$\underline{x} = \underline{P} \cdot \underline{a}$$

$$\begin{bmatrix} \beta_{11} & \beta_{12} \\ \beta_{21} & \beta_{22} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$



$$x_2 = \cos \alpha \cdot a_2 - \sin \alpha \cdot a_1$$

$$x_1 = \sin \alpha \cdot a_2 + \cos \alpha \cdot a_1$$

$$\alpha = 45^\circ$$

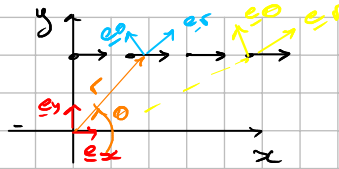
$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

$$\cos \alpha = \frac{\sqrt{2}}{2}$$

$$\sin \alpha = \frac{\sqrt{2}}{2}$$

$$u = 1$$

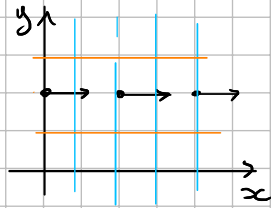
$$v = 0$$



$$u_r = v \cdot \cos \theta - u \cdot \sin \theta = -\sin \theta$$

$$u_\theta = v \cdot \sin \theta + u \cdot \cos \theta = \cos \theta$$

$$\underline{v} = u_r \cdot \underline{e}_r + u_\theta \cdot \underline{e}_\theta$$



$$(u_r, u_\theta)$$



$$\text{quiver} \rightarrow (u, v)$$